Domain-Specific Foundation Model for Liver Transplant

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BACKGROUND

- Liver transplantation involves uniquely complex donor-recipient, longitudinal and multimodal clinical decision making
- Foundation Models (FMs) such as GPT are rapidly transforming healthcare
- Liver transplantation, however, remains underexplored with most models focused on isolated tasks

There is a need to move beyond narrow task specific models towards domain-specific liver transplant foundation models

DATASET CURATION AND PREPROCESSING

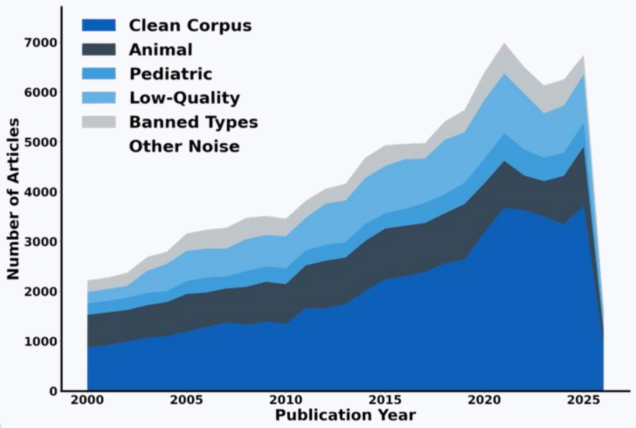

Fig 1. Data Preprocessing

Preprocessing Summary

- Q1/Q2 quality filtering
- MeSH filtering
- Structural extraction
- Deduplication

MODEL DEVELOPMENT

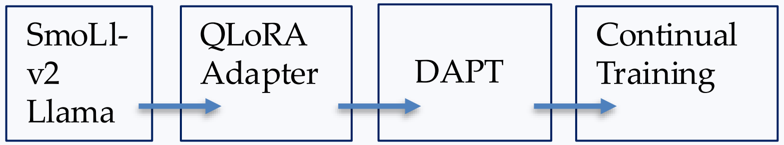

Fig 4. Modeling Pipeline

Specialized Liver Model

- QLoRA-based parameter efficient adaptation
- One training epoch
- Sequence packing and diagonal masking
- Casual Language Modeling

LIVER TRANSPLANT SPECIFIC QA

Question	Base Model	Our Model
Tacrolimus ID	FK-508	FK-506
MELD	Original MELD	MELD 3.0
HCC Eligibility	Missed Milan Criteria	Milan Criteria

Table 1. Qualitative Comparison

The specialized model produced liver responses:

- More up-to date Accurate and clinically grounded
- Fewer hallucination

EVALUATION AND RESULTS

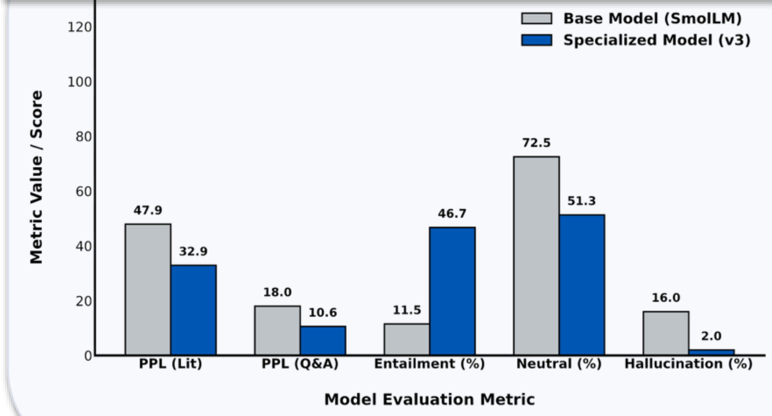

Fig 3. Quantitative Comparison

We developed a liver-transplant focused clinical QA benchmark using:

- OPTN, AASLD, textbooks, Cleavland clinic resources

Specialized Model showed:

- Reduced perplexity
- Reduced hallucinations
- Increased factual entailment

CONCLUSION

- We present the first liver-transplant foundation model
- This work lays the foundation for next-generation FMs that are clinically useful and trustworthy

FUTURE WORK

- Incorporate De-identified clinical notes
- Scale to larger Llama Models
- Incorporate RAG for protocols
- Compare to larger models (GPT)